ONE WATER OPERATING SYSTEM

One Water AI Executive Fellowship

Master curriculum, program value, participant experience, and implementation portfolio

MASTER CURRICULUM CANDIDATE | AUGUST 2026

Built for water, wastewater, stormwater, and One Water leadership

Powered by APAS.AI

THE PROBLEM WE ARE SOLVING

AI fluency is not the finish line.

The water sector needs professionals who can separate a demonstration from a defensible utility application. The work begins with technology, but it has to reach data, knowledge, governance, people, economics, and accountable decisions.

WHAT THE PARTICIPANT LEARNS

Explain the technology, test the evidence, identify useful applications, design controls, and lead a bounded pilot.

WHAT THE ORGANIZATION RECEIVES

A prioritized opportunity portfolio, governance record, investment case, 90-day pilot brief, and one-year roadmap.

THE PROMISE

Move one meaningful utility opportunity from enthusiasm to a source-backed, governed, measurable pilot that leadership can understand, challenge, fund, or stop.

SIX MONTHS OF APPLIED LEARNING

Eight courses. Sixty-four modules. One implementation portfolio.

24	132	25	$10K
WEEKS	GUIDED HOURS	PARTICIPANTS PER COHORT	TARGET TUITION

LEARNING SYSTEM

01 Articulate modules

02 Live executive forums

03 Applied studios

04 OWOS knowledge graph

05 Fieldbook

06 Capstone defense

A SELECTED PROFESSIONAL COHORT

Built for people who influence consequential decisions.

- General managers, executive directors, and deputy directors
- Operations, maintenance, engineering, and water-quality leaders
- Capital, asset, customer, finance, and communications leaders
- Data, technology, cybersecurity, innovation, and transformation officers
- Legal, privacy, procurement, risk, audit, and governance professionals
- Senior consultants, solution providers, researchers, regulators, and nonprofit leaders
- Selected emerging leaders with sponsor support and a real application context

ADMISSIONS EXPECTATION

Programming is not required. Participants need a real professional context, organizational support, and the authority or influence to complete an applied capstone.

THE SIX-MONTH SEQUENCE

From shared language to a governed pilot.

01 **AI Fluency for One Water**
Weeks 1 to 3 | Modules 1 to 8

02 **Data, Knowledge, and Trusted Context**
Weeks 4 to 6 | Modules 9 to 16

03 **Agents, Skills, and Orchestration**
Weeks 7 to 9 | Modules 17 to 24

04 **Utility Applications and Opportunity Design**
Weeks 10 to 12 | Modules 25 to 32

05 **Governance, Security, and Human Authority**
Weeks 13 to 15 | Modules 33 to 40

06 **Strategy, Economics, and Organizational Change**
Weeks 16 to 18 | Modules 41 to 48

07 **Applied AI Studios**
Weeks 19 to 21 | Modules 49 to 56

08 **Capstone, From Opportunity to Governed Pilot**
Weeks 22 to 24 | Modules 57 to 64

WEEKS 1 TO 3

AI Fluency for One Water

Build a shared language for AI and connect every term to a recognizable utility task, decision, or consequence.

EIGHT APPLIED MODULES

01 **AI and utility work**
Keep: AI use-case boundary card

02 **How we arrived here**
Keep: AI family map

03 **Models, tokens, prompts, and context**
Keep: prompt and context anatomy sheet

04 **Language, images, audio, and multimodal AI**
Keep: multimodal utility opportunity card

05 **Chatbot, retrieval, workflow, agent, or agentic system**
Keep: system classifier

06 **Working effectively with an AI assistant**
Keep: reusable utility prompt pattern

07 **Claude Skills and reusable organizational instructions**
Keep: first utility Claude Skill specification

08 **Fluency checkpoint**
Keep: AI terms and decision field card

WEEKS 4 TO 6

Data, Knowledge, and Trusted Context

Understand why useful AI depends on connected records, shared meaning, current sources, permission, and traceable evidence.

EIGHT APPLIED MODULES

09 **Data before AI**
Keep: readiness diagnosis

10 **Data quality and observability**
Keep: data-quality control card

11 **Metadata, lineage, and provenance**
Keep: lineage trace

12 **Taxonomy and ontology**
Keep: utility concept map

13 **Knowledge graphs and the systems lens**
Keep: knowledge relationship map

14 **Embeddings and vector retrieval**
Keep: retrieval comparison sheet

15 **Retrieval-augmented generation**
Keep: RAG answer contract

16 **Context engineering and the trusted answer**
Keep: data and knowledge readiness map

WEEKS 7 TO 9

Agents, Skills, and Orchestration

See how AI moves from a response to bounded work, then design the instructions, tools, handoffs, permissions, and stop conditions required to keep people in control.

EIGHT APPLIED MODULES

17 **Before the agent**
Keep: architecture decision record

18 **Inside the agent loop**
Keep: agent-loop trace

19 **Agent anatomy**
Keep: dependency map

20 **Tools, connectors, and the Model Context Protocol**
Keep: tool and connector register

21 **Skills as organizational recipes**
Keep: working Claude Skill package

22 **Memory, state, and durable records**
Keep: state and record plan

23 **Orchestration and multi-agent handoffs**
Keep: handoff contract

24 **Design the smallest useful agent**
Keep: Utility Agent Canvas

WEEKS 10 TO 12

Utility Applications and Opportunity Design

Move from generic AI excitement to a disciplined portfolio of opportunities grounded in utility work, available evidence, affected people, and measurable value.

EIGHT APPLIED MODULES

25 **Operations and operator support**
Keep: operations opportunity card

26 **Maintenance and asset management**
Keep: maintenance opportunity card

27 **Water quality, laboratory, and compliance support**
Keep: evidence-chain card

28 **Customer service and public communication**
Keep: customer-service use-case card

29 **Capital planning, engineering, and project delivery**
Keep: capital-delivery opportunity card

30 **Stormwater, wastewater, drinking water, and One Water coordination**
Keep: One Water relationship map

31 **Finance, procurement, legal, and administration**
Keep: administrative opportunity card

32 **Opportunity portfolio and prioritization**
Keep: prioritized utility opportunity portfolio

WEEKS 13 TO 15

Governance, Security, and Human Authority

Build governance into the design so that the organization knows what the system may do, what it may not do, who checks it, and what evidence survives.

EIGHT APPLIED MODULES

33 **Risk and consequence tiers**
Keep: use-case risk tier

34 **Privacy, confidentiality, and data rights**
Keep: privacy and data-use record

35 **Cybersecurity and adversarial failure**
Keep: threat and control map

36 **Identity, permissions, and segregation of duties**
Keep: permission matrix

37 **Human authority, escalation, and stop conditions**
Keep: authority and escalation map

38 **Evaluation, testing, and continuous assurance**
Keep: evaluation plan

39 **Vendor, model, and procurement governance**
Keep: AI procurement scorecard

40 **The governed AI application record**
Keep: AI governance and control record

WEEKS 16 TO 18

Strategy, Economics, and Organizational Change

Turn isolated experiments into a sequenced organizational program with value measures, ownership, funding, workforce preparation, and an operating rhythm.

EIGHT APPLIED MODULES

41 **Problem formulation before technology selection**
Keep: problem statement

42 **Value, baseline, and measurement**
Keep: value baseline

43 **The full cost of AI**
Keep: total-cost model

44 **Build, buy, partner, or do nothing**
Keep: sourcing decision record

45 **AI operating model and decision rights**
Keep: operating-model canvas

46 **Workforce capability and responsible adoption**
Keep: workforce capability plan

47 **Portfolio governance and sequencing**
Keep: portfolio roadmap

48 **Executive narrative and one-year roadmap**
Keep: investment case and one-year roadmap

WEEKS 19 TO 21

Applied AI Studios

Build enough to understand the mechanics, expose failure, and make better implementation decisions. No programming background is required.

EIGHT APPLIED MODULES

49 **Prompt and context studio**
Keep: tested prompt pattern

50 **Source-grounded answer studio**
Keep: RAG test record

51 **Claude Skill studio**
Keep: tested Claude Skill

52 **Tool-use and permission studio**
Keep: tool-use evidence log

53 **Agent workflow studio**
Keep: agent trace

54 **Multi-agent decision-room studio**
Keep: multi-agent decision record

55 **Failure and red-team studio**
Keep: failure and recovery log

56 **Evaluation and pilot rehearsal**
Keep: prototype evidence package

WEEKS 22 TO 24

Capstone, From Opportunity to Governed Pilot

Convert one real organizational opportunity into a pilot that leadership can understand, challenge, measure, govern, and stop when necessary.

EIGHT APPLIED MODULES

57 **Capstone charter and stakeholder agreement**
Keep: capstone charter

58 **Current-state process and baseline**
Keep: current-state map

59 **Data and knowledge readiness**
Keep: readiness plan

60 **Architecture and workflow decision**
Keep: architecture decision record

61 **Governance, security, and human authority**
Keep: pilot control plan

62 **Evaluation, economics, and success criteria**
Keep: measurement and financial plan

63 **Ninety-day pilot and one-year adoption roadmap**
Keep: 90-day pilot brief and one-year roadmap

64 **Capstone defense and professional commitment**
Keep: defended implementation portfolio

THE COMPANION PRODUCT

Learning that accumulates into professional work.

The One Water AI Fieldbook is part workbook, part laboratory notebook, part decision record, and part executive playbook. Each module adds one useful artifact. At graduation, those artifacts form a complete implementation portfolio.

RECORD 1

AI use-case boundary and terms field card

RECORD 2

Data and knowledge readiness map

RECORD 3

Claude Skill package and Utility Agent Canvas

RECORD 4

Prioritized utility opportunity portfolio

RECORD 5

Governance, permissions, evaluation, and procurement records

RECORD 6

Investment case, workforce plan, and operating model

RECORD 7

Tested prototype evidence package

RECORD 8

90-day pilot brief and one-year roadmap

THE PROOF OF LEARNING

A pilot that can be understood, challenged, measured, and stopped.

01	Problem, current process, affected people, and baseline
02	Accountable sponsor, owner, reviewers, users, and decision rights
03	Approved sources, data limitations, definitions, and provenance
04	Architecture decision and reason simpler alternatives were rejected
05	Permissions, controls, review, incident, and escalation design
06	Historical test cases, thresholds, cost limits, and human-review effort
07	Ninety-day pilot sequence and one-year adoption roadmap
08	Candid statement of what remains unknown

WHAT IS BUILT, AND WHAT REMAINS

The blueprint is complete. The fellowship still has to earn release.

BUILT NOW

A 24-week architecture, 64-module curriculum, premium value proposition, Fieldbook system, applied capstone, free-resource strategy, source controls, and the first module authoring package.

REQUIRED BEFORE ENROLLMENT

Faculty and practitioner review, source rights, 64 Articulate lessons, live-studio plans, the completed Fieldbook, assessment and credential controls, design-cohort evidence, accessibility review, and delivery integration.

CREDENTIAL BOUNDARY

Completion does not authorize engineering, operations, regulation, finance, procurement, or security decisions. Any fellowship credential remains disabled until identity, evidence, assessment, review, and issuance gates are independently approved.

CURRICULUM COMPARISON SOURCES

External programs informed the gap assessment, not the utility content.

MIT SLOAN AI EXECUTIVE ACADEMY

https://executive.mit.edu/ai-executive-academy.html

MIT SLOAN AI ESSENTIALS

https://executive.mit.edu/ai-essentials.html

ANTHROPIC: THE COMPLETE GUIDE TO BUILDING SKILLS FOR CLAUDE

https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf

SOURCE BOUNDARY

These public references support curriculum comparison only. They do not provide authority for utility operations, and no third-party course content is reproduced in this document.